

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

TRAVEL MEDICINE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE	Adverse Effects
AMS	Acute Mountain Sickness
ASA	Aspirin
CATMAT	Committee to Advise on Tropical Medicine and Travel
CDC	Centre for Disease Control and Prevention
CI	Contraindicated
D/C	Discontinuation
DEET	N,N-Diethyl-meta-toluamide
DOAC	Direct Oral Anticoagulant
GI	Gastrointestinal
HACE	High-Altitude Cerebral Edema

HAPE	High-Altitude Pulmonary Edema
ISTM	International Society of Travel Medicine
LMWH	Low Molecular Weight Heparin
N/V	Nausea/Vomiting
PE	Pulmonary Embolism
PHAC	Public Health Agency of Canada
QHS	Every Night At Bedtime
Tx	Treatment
UV	Ultraviolet
VTE	Venous Thromboembolism
WHO	World Health Organization

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Pre-travel Consultation

Traveler's Risk Assessment

Prevention Strategies and Recommendations

- High Altitude Illnesses
- Insect Protectants
- Water and Food Safety
- Thrombosis Prevention
- Jet Lag
- Vaccinations

LEARNING OBJECTIVES

- 1 Perform a risk assessment for traveling patients
- 2 Educate patients on useful resources to ensure safe and enjoyable travel
- 3 Discuss prevention strategies for patients traveling to high-risk areas
- 4 Discuss treatment strategies for specific travel-related conditions
- 5 Recommend vaccinations for high-risk patients traveling to endemic areas
- 6 Understand prevention and treatment strategies for traveler's diarrhea

OVERVIEW OF THE TRAVEL PROCESS



PRE-TRAVEL CONSULTATION

Patients should see a travel specialist at least **6 weeks** prior to travel



Clinician will review patient's **medical and immunization history**



Clinician will discuss any trip-specific health concerns based on patient's **medical history and area of travel**

PRE-TRAVEL CONSULTATION

Health Background



1. **Past medical history** (age, underlying conditions, allergies, medications)
2. **Special conditions** (pregnancy, breastfeeding, disability, recent surgery, immunocompromised)
3. **Immunization history** (routine vaccines and travel vaccines)
4. **Prior travel experience** (past malaria prevention, altitude trips)



Trip Information



1. **Itinerary** (countries, regions, rural vs. urban)
2. **Timing** (duration of trip, season of travel, time until departure)
3. **Reason for travel** (business, tourism, friends)
4. **Special activities** (diving, medical care, camping, hiking)

PRE-TRAVEL RESOURCES



Public Health Agency of Canada

- Information on outbreaks
- List of clinics providing travel health services in Canada



Centre for Disease Control and Prevention

- CDC Yellow Book information for international travel
- US recommendations
- International travel health information for health professionals



World Health Organization

- General travel advice
- Country-specific malaria risk
- Antimalarial recommendations



International Society of Travel Medicine

- Online community of travel health providers globally
- Offers a certificate in travel health



Government of Canada

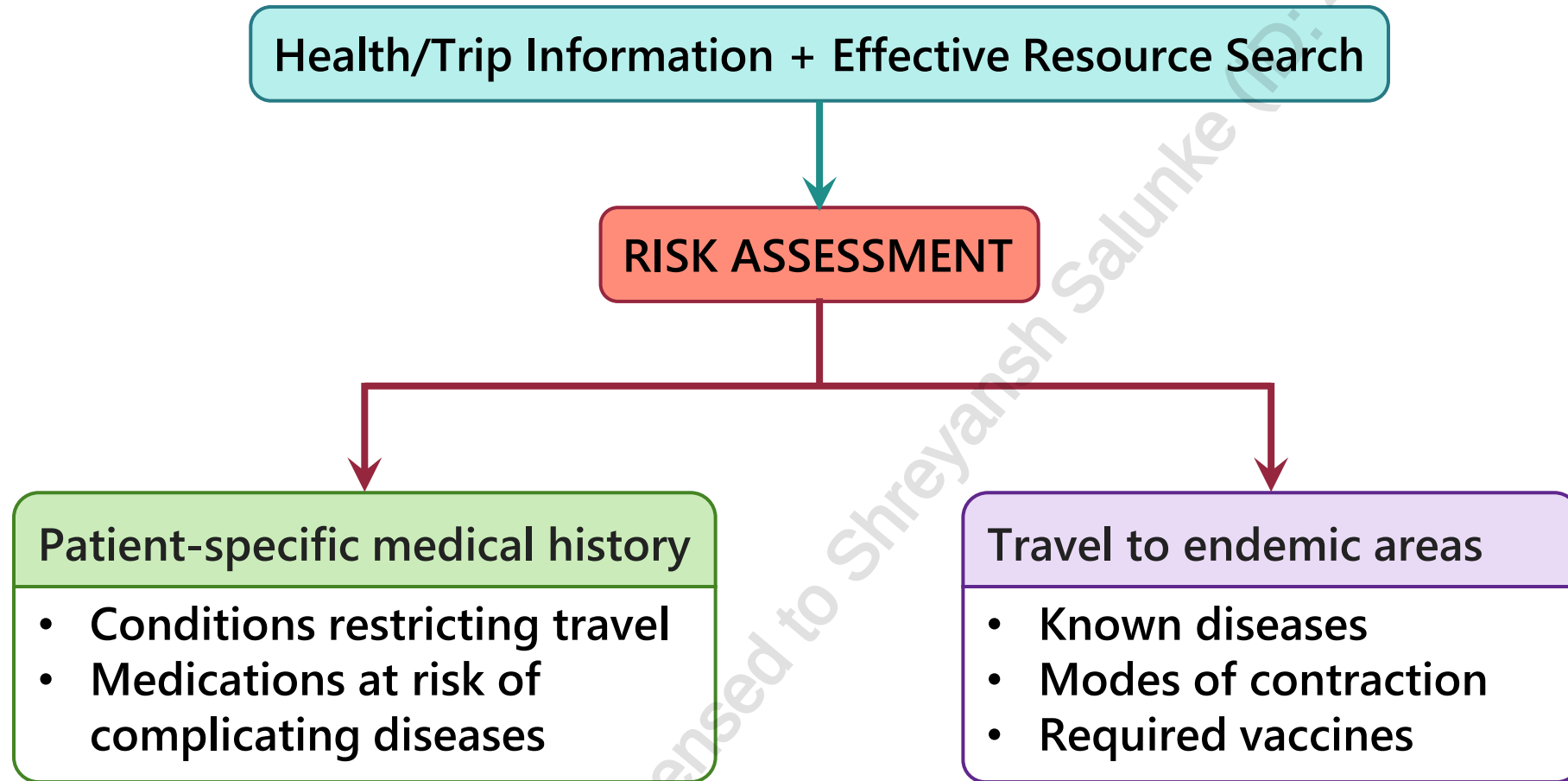
- Travel and tourism advice for Canadians



Committee to Advise on Tropical Medicine and Travel



RISK ASSESSMENT



PREVENTION STRATEGIES

High Altitude Illnesses

Insect Protection

Sun Safety

Water and Food Safety

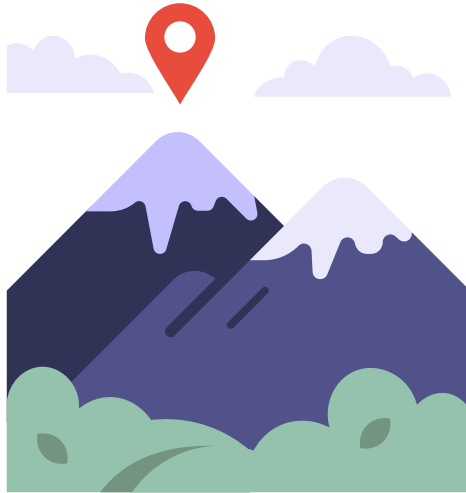
Thrombosis Prevention

Jet Lag

Vaccinations

HIGH ALTITUDE ILLNESS

- For patients travelling to regions with **high elevations**
- Ask patients about travel plans, specifically if any regions are higher than **2000 meters above sea level**



HIGH ALTITUDE ILLNESS

Acute Mountain Sickness (AMS)

Symptoms (**within 48h of new altitude**):

- CNS (headache, lightheadedness)
- GI (anorexia)
- MSK (weakness, fatigue)

High Altitude Pulmonary Edema (HAPE)

Symptoms include:

- Respiratory (dyspnea, cough, congestion)
- Pulmonary sounds (crackles, wheezing)
- CV (tachycardia, chest tightness, cyanosis)

HAPE can be more rapidly fatal than HACE

HAPE is not necessarily a progression from AMS - it can occur on its own

High Altitude Cerebral Edema (HACE)

Symptoms of AMS along with what seems like alcohol intoxication

- Ataxia
- Changes in mental status
- Severe headache
- Death can occur within 24 hours of development if person fails to descend

HACE (uncommon) is of the end stage of AMS and normally occurs **within three days**
Most often associated with HAPE

HIGH ALTITUDE ILLNESS

Prevention

- **Ascend slowly** and avoid direct transport to altitudes >3000 meters
- Spend 1 or 2 nights at 2500 – 3000 meters
- **>2750-3000 meters, sleeping altitude should not be increased by more than 300-500 meters/night** (rest 2 nights at the same altitude every 2-3 days)
- **Avoid alcohol/sedatives/hypnotics**
- High carb diet ↓ AMS symptoms by 30%
- Avoid overexertion

Prophylactic Options for Rapid Ascents

(<24 hours and >3000 meters or prior history of AMS/HAPE/HACE)

- **Acetazolamide 125 mg Q12H** beginning the day before ascent until 48 hours after highest altitude reached or descent initiated
- **Dexamethasone 4 mg Q12H**

Treatment of AMS Symptoms

- **Acetazolamide 250mg Q12H** (initiated soon after AMS symptoms) and continue until symptoms have cleared (to a max of 4 days)
- **Rest and acclimatization at one altitude**
- **Dexamethasone 4 mg Q6H** until 24 hours after symptoms resolve or descent (use generally limited to 48-72 hours)
- Simple analgesics like ibuprofen or acetaminophen for headache

INSECT PROTECTION

- Always **consult the Public Health Agency of Canada (PHAC)** to evaluate the risk of insect-borne illnesses that are endemic to the area
- Make patient aware of the risks in the area and address insect prevention and treatment strategies



INSECT PROTECTION: METHODS OF PREVENTION

Avoidance

- Recognize that there are two types of insects
- Aedes = mosquitos active during the **daytime** (e.g. dengue fever, zika)
- Anopheles = mosquitos active during the **night** (e.g. malaria)

Physical Barriers

- Use **screens** around the beds and room (permethrin-treated recommended)
- Screens (mesh size <1.5mm) over the windows
- **Wear shoes, socks and full clothing during peak hours**

Chemical Barriers

- DEET: **30% TID for adults > 12 years**
- DEET: **10% TID for children aged 2-12 years**
- DEET: **10% daily for children 6 month to 2 years**
- Icaridin: 20% BID for children > 6 months
- **Mosquito net: < 6 months**



MALARIA

Overview

- Caused by a protozoan parasite of the *Plasmodium* genus—transmitted by mosquitos
- Each genotype can infect humans, some more serious than others (*P. falciparum malarium*)
 - Five different species of plasmodium: *falciparum*, *malariae*, *ovale*, *vivax*, *knowlesi*
- Risks of acquiring malaria are **endemic** to specific regions across the globe, always check CDC if a traveler is heading to a highly endemic area

Clinical Presentation



Flu-like symptoms:

Fever, chills, sweating, headache, myalgia



GI symptoms:

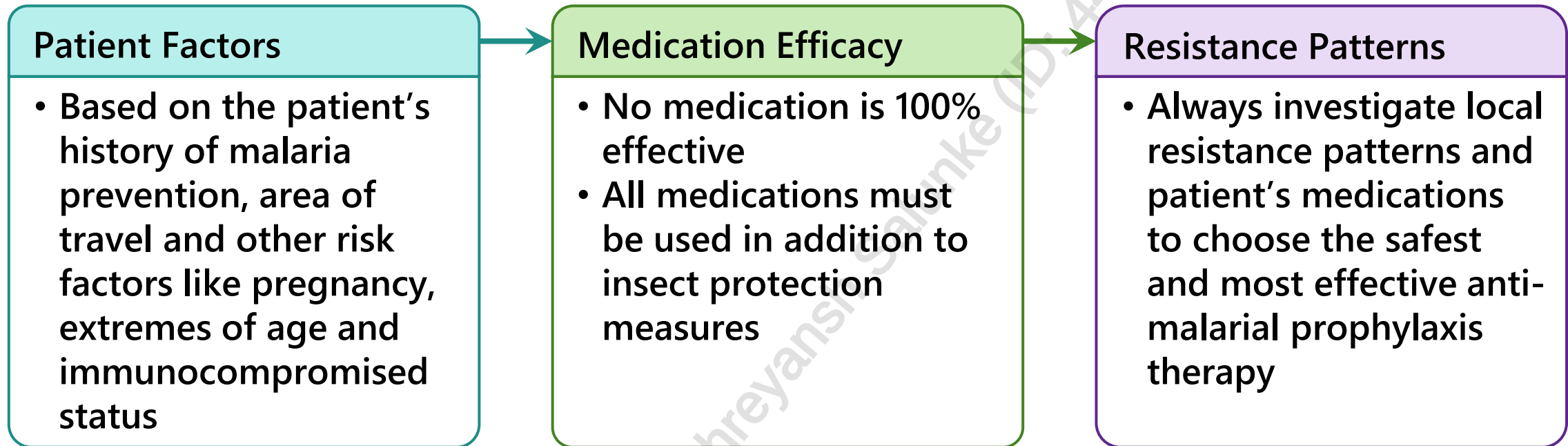
Nausea/vomiting



Severe symptoms of *P. falciparum malarium*:

jaundice, abnormal bleeding, convulsions (leading to shock, kidney failure, even death)

MALARIA PROPHYLAXIS



MALARIA PHARMACOLOGICAL THERAPY 1/5

Atovaquone-Proguanil



- 250/100 mg daily (adults)
- 62.5/25 mg (dosed by weight, given to children >5 kg)
- Take with fatty food (or milky drink) **1-2 days before travel** to endemic area **and daily while in risk area**
- **Continue for 7 days after leaving the malaria risk area**
- Well tolerated (mild abdominal pain, headache, N/V, insomnia)

Pros of Treatment

1. Good for last minute travelers (only need to initiate 1-2 days before travel)
2. Good for short trips (only require 7 days of post-travel treatment)
3. Pediatric formulations available

Cons of Treatment

1. More expensive than other options
2. **Avoid in pregnancy or breastfeeding a child <5 kg**
3. **Avoid in renal impairment (<30 ml/min)**

MALARIA PHARMACOLOGICAL THERAPY 2/5

Primaquine



- Use in *P. vivax*-endemic areas, PART (presumptive anti-relapse tx), or alternative agent in chloroquine-resistant areas (5-10% less effective)
- 30 mg tablet daily
- Begin **1-2 days before travel and continue for 7 days after end of trip**
- Well-tolerated (GI upset), methemoglobinemia (rare)

Pros of Treatment

1. Good for last-minute travelers (only need to initiate 1-2 days before travel)
2. Good for short trips (only require 7 days of post-travel treatment)
3. Most effective for *P. vivax* (drug of choice where this strain is endemic)

Cons of Treatment

1. **Contraindicated in severe G6PD deficiency** (cost and delays to being tested for this deficiency)
2. **Avoid in pregnancy/breastfeeding**
3. Gastrointestinal AE often leads to d/c of therapy – best to take with food

MALARIA PHARMACOLOGICAL THERAPY 3/5

Hydroxychloroquine



- Useful when heading to area without known chloroquine resistance
- **Start 1 week before** entering malaria risk area
- Take **one tablet weekly throughout travel and continue for 4 weeks after leaving malaria risk area**
- 400 mg weekly: adults
- 6.5 mg/kg weekly: children
- Side effects: GI upset, headache, blurred vision, pruritus

Pros of Treatment

1. Good for long trips (patients only require weekly dose)
2. Safe in pregnancy/breastfeeding: drug of choice

Cons of Treatment

1. Many endemic regions are known to be chloroquine-resistant
2. Patients must take the medication for 4 weeks post-travel
3. Not good for last minute travelers (must be started 1 weeks prior to travel)
4. **Avoid in patients with psoriasis or seizures**
5. Potential caution in those with G6PD deficiency d/t risk of hemolytic anemia

MALARIA PHARMACOLOGICAL THERAPY 4/5

Doxycycline



- 100 mg daily
- Begin **1-2 days before** traveling and continue for **28 days after leaving** malaria risk area
- Side effects: photosensitivity, N/V
- Take with food and +++ water while remaining in upright position to avoid esophageal ulceration

Pros of Treatment

1. Good for last minute travelers (only need to initiate 1-2 days before travel)
2. Least expensive antimalarial medication

Cons of Treatment

1. Risks of photosensitivity
2. **Contraindicated in pregnancy, breastfeeding, or children under 8 years old**
3. Must take for 28 days after leaving malaria risk area

MALARIA PHARMACOLOGICAL THERAPY 5/5

Mefloquine



- Use in areas sensitive to this medication
- 250 mg tablet weekly
- Begin **1 week before travel and continue for 4 weeks after return**
- First time users recommended to start 2-3 weeks prior to leaving if time permits to assess for AEs
- **CI in many psychiatric disorders (depression, seizures, anxiety)**

Pros of Treatment

1. Good for long trips (patients only require weekly dose)
2. Can be initiated within 1 week of travel
3. Safe in pregnancy (DOC if chloroquine-resistant area)
4. Weight-based dosing for pediatric patients

Cons of Treatment

1. Areas of mefloquine resistance
2. Issues with this medication when patient has history of seizures and psychosis
3. Must take for 4 weeks after travel
4. Known to have severe side effects leading to discontinuation

MALARIA PROTECTION SUMMARY

Agent	Indication	Use in Pregnancy
Atovaquone-Proguanil	Start 1-2 days before travel, continue while in area, and for 7 days after leaving	No
Primaquine	Start 1-2 days before travel, continue while in area, and continue for 7 days after leaving	No
Hydroxychloroquine	Start 1 week before travel (dosed weekly), continue while in area, and continue for 4 weeks after leaving	Yes
Doxycycline	Start 1-2 days before traveling, continue while in area, and for 28 days after leaving	No
Mefloquine	Start 1 week before travel and, continue while in area, and continue for 4 weeks after return	Yes



DENGUE FEVER

DENGUE FEVER

- An acute febrile illness caused by virus transmitted by Aedes species mosquitos in endemic areas or exposure to dengue infected blood
- Common in more than 100 countries
- Prevention is key (use insect repellents and wear tight clothing)
- Most common symptom is fever with GI symptoms, rash or aches/pains
- 1 in 20 individuals with dengue will develop severe dengue
- Supportive care

Clinical Presentation



Headache



Rash (2 days after fever)



Joint pain



- Symptoms* begin 4-7 days after exposure
- Seldom causes death

**some travelers are asymptomatic*



CHIKUNGUNYA

CHIKUNGUNYA

- A disease caused by virus transmitted by *Aedes* mosquitos in endemic areas
- There is a vaccine available to prevent chikungunya virus
- Prevention is key (use insect repellents and wear tight clothing)
- Most common symptoms are fever and joint pain
- Supportive care: acetaminophen for fever and muscle pain, rest, fluids

Clinical Presentation



Headache



Rash



Joint pain (can persist for years causing chronic disability)



- Symptoms* begin 3-12 days after exposure
- Mimics dengue fever

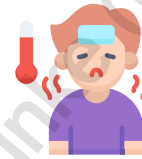
**some travelers are asymptomatic*



ZIKA

- A disease caused by virus transmitted by Aedes mosquitos in endemic areas
- No specific treatment or vaccine!
- Zika can be passed from a pregnant woman to her fetus
- Pregnant women should avoid areas
- Avoid conception x 2 months (woman traveled to high risk/outbreak area) and x 3 months (man traveled to high risk/outbreak area) after leaving affected area

Clinical Presentation*



Fever



Rash



Joint pain (can persist for years causing chronic disability)



Can cause microencephaly and congenital brain abnormalities in fetus

**some travelers are asymptomatic*

WATER AND FOOD SAFETY

- Educate patients on the importance of water and food safety while travelling
- **Risk of neglecting recommendations → traveler's diarrhea**
- Assess the risk of the area by looking at endemic pathogens through PHAC
- Educate patients on food and water safety especially those that are pregnant, young, old, and immunocompromised

Low Risk Foods



Food that is fully cooked



Hard-boiled eggs



Clean and peeled fruits and veggies



Dairy products that have been pasteurized



Dry foods (e.g. crackers)



Sealed beverages (e.g. canned soda)



Boiled water or water treated for contaminants

High Risk Foods



Rare meats served at room temperature



Raw and unwashed fruits or veggies



Tap/well water (possibly untreated for contaminants)



Fountain drinks



Dairy products that have NOT been pasteurized



Salads



Raw or runny eggs



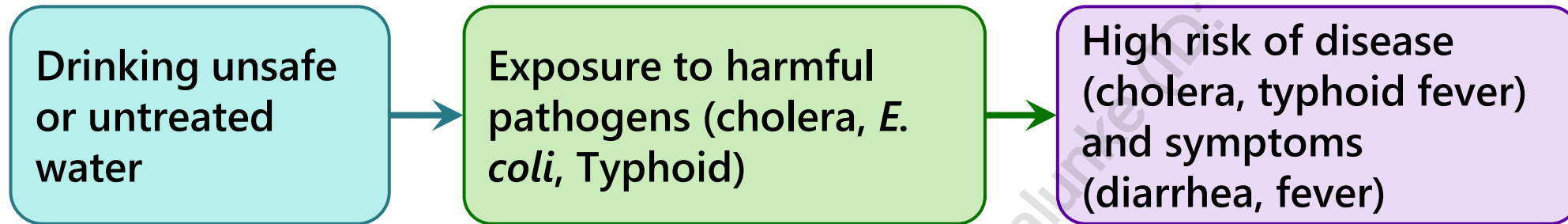
Bushmeat (e.g. monkey or bats)



Food from street vendors

Remember to boil, cook, and peel your food!

WATER AND FOOD SAFETY



WATER AND FOOD SAFETY

CHLORINE BLEACH

- The least reliable
- Add 2 drops of chlorine household bleach to 1 L of water and let sit for 30 minutes

IODINATION

- Recommended only for a short term (< 2 weeks) if unable to boil
- **Contraindicated** in pregnant women and children or thyroid disease patients
- Tincture of 2% iodine (5 drops/L then let stand for 30 minutes)

BOILING WATER

- **Most reliable and preferred method**

THROMBOSIS PREVENTION

- Address the risks of thrombosis for long flights, as well as the risks of thrombosis following the flight
- Educate patients at high risk of traveler's thrombosis regarding proper prevention
- **Anyone traveling more than 4 hours by air, car, or bus may be at risk of blood clot (CDC)**
- Patients with a history of DVT/PE due flying will require evaluation by a physician regarding anticoagulant therapy for flights **over 4 hours**



THROMBOSIS PREVENTION

Risk Factors For VTE

- Duration of travel (8 hours have strongest association)
- Previous patient risks (VTE, cancer, pregnancy, age, surgery, injury, clotting)
- Medications (oral contraceptives, hormone replacement therapy)
- Immobility on the flight
- Greatest risk first 2 weeks after travel, then slowly back to baseline by 8 weeks
- Obesity (BMI ≥ 30)
- Window seating is associated with a greater risk

Non-Pharmacological Prevention

- Stay hydrated and **avoid alcohol** on the flight
- Contract muscles in calves regularly or get up often to walk around (every 1-2 hours)
- **Below-knee compression stockings** (20-30 mmHg of pressure on ankles) are not routinely recommended for those with no risk factors (recommended for long distance travelers at an increased risk of VTE)
- Avoid constrictive clothing
- Improve leg mobility – minimize baggage under the seat

Pharmacological Prevention

- ASA **does not** appear to reduce risk of travel-related blood clots or embolisms
- **LMWH or DOAC** may be recommended after physician assessment for long distance travelers with baseline risk factors for VTE
- If no increased risk for VTE, prophylactic anticoagulant or antiplatelet not routinely recommended
- If already on prophylactic or treatment anticoagulant or antiplatelet, no additional treatment recommended

JET LAG

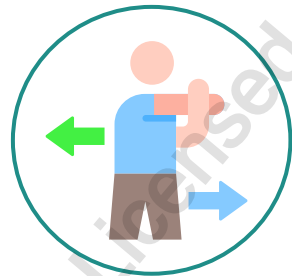
Clinical Presentation

- Malaise
 - Fatigue
 - Disruption of sleep-wake cycles
 - Irritability and impaired cognitive abilities
-
- Travelling eastward is more severe than westward
 - Takes 1 day for every hour of time difference to acclimatize – suggested guideline

JET LAG

Non-Pharmacological Prevention

- Adjust sleep patterns to destination prior to departure
- Stay well-hydrated during flight; avoid alcohol; avoid large meals
- Remain active during flight by stretching or walking (also helps to stay awake to adjust to new time zone)



Pharmacological Prevention

- **Melatonin** is produced endogenously at nighttime by pineal gland to assist body with sleep
- Most well-known treatment for jet lag
- Exogenous melatonin is effective in reducing jet lag symptoms
- Melatonin dosing: 3-5 mg QHS – IR formulation is most effective, take first night after arrival (before nocturnal sleep in the new time zone)
- AE: abdominal cramps, dizziness, headache, irritability, drowsiness



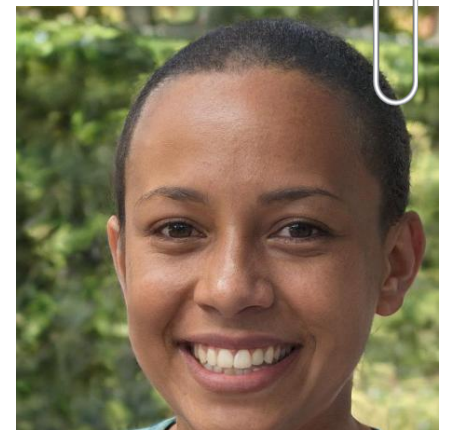
OVERVIEW OF TRAVEL MEDICINE PROCESS



CASE

SR is a 34-year-old female who is planning a two-week trip to Ghana, where she will be visiting family in Accra and spending time in rural areas. She has no significant medical history, and she only takes a multivitamin daily. She has no known drug allergies. As part of her trip, she will be engaging in various activities, including sightseeing, outdoor excursions, and visiting local markets, potentially increasing her exposure to mosquito bites. She is aware that malaria is endemic in Ghana, with the most common species being *P. falciparum*. She has read about different prevention options and would prefer a shorter course of medication. SR leaves for her trip in less than one week and is eager to ensure she is adequately prepared.

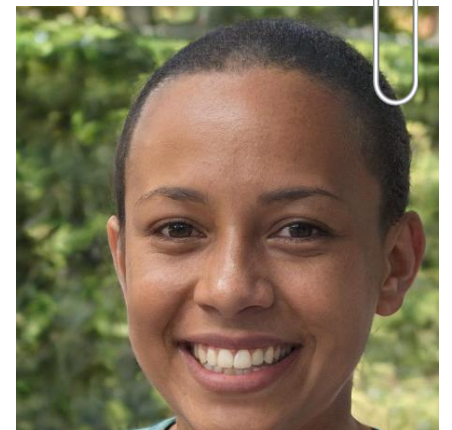
1. Which of the following counselling points is the most important to tell SR to reduce her risk of malaria while travelling?
 - a) Wearing long-sleeved clothing and using insect repellent
 - b) Drinking only bottled or filtered water to avoid gastrointestinal infections
 - c) Getting the yellow fever vaccine before travel
 - d) Avoiding eating food from street vendors



CASE

SR is a 34-year-old female who is planning a two-week trip to Ghana, where she will be visiting family in Accra and spending time in rural areas. She has no significant medical history, and she only takes a multivitamin daily. She has no known drug allergies. As part of her trip, she will be engaging in various activities, including sightseeing, outdoor excursions, and visiting local markets, potentially increasing her exposure to mosquito bites. She is aware that malaria is endemic in Ghana, with the most common species being *P. falciparum*. She has read about different prevention options and would prefer a shorter course of medication. SR leaves for her trip in less than one week and is eager to ensure she is adequately prepared.

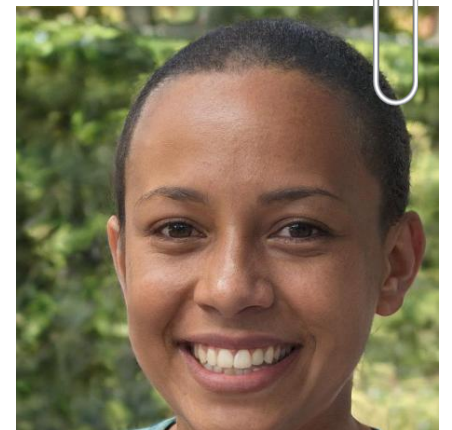
2. Which of the following is the most appropriate malaria prophylaxis option for SR?
- a) Atovaquone-Proguanil 250/100 mg daily
 - b) Primaquine 30 mg daily
 - c) Mefloquine 250 mg weekly
 - d) Doxycycline 100 mg daily



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3. Which of the following options would be safe to use if SR was pregnant?
- a) Primaquine
 - b) Doxycycline
 - c) Atovaquone-Proguanil
 - d) Mefloquine



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CHANGE LOG

July 2025

- Spelling, grammar and formatting changes made throughout

September 2025

- Slide 31: Changed 2 hours to 4 hours
- Slide 32: Added to risk factors and non-pharmacological prevention

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 13: Minor updates to HACE
- Minor formatting edits made throughout